

Design and Characterization of a 32 GHz Differential Silicon Photonic Mach–Zehnder Modulator for Radio-Over-Fiber Applications

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Abstract—Next-generation 5G and beyond wireless communication systems demand fronthaul architectures that support high bandwidth, low latency, scalability, and energy efficiency. Analog Radio-over-Fiber (RoF) has emerged as a promising solution by enabling centralized baseband processing and efficient transport of mm-wave signals over optical fiber. In this work, a 32 GHz differential traveling-wave silicon photonic Mach–Zehnder modulator (MZM) is designed, fabricated, and characterized for analog RoF applications. The proposed modulator employs PN-depletion phase shifters and a slow-wave electrode architecture to achieve wideband operation, high linearity, and CMOS compatibility. Experimental results demonstrate electrical–electrical bandwidths approaching 38.8 GHz and successful integration with a 16–32 GHz CMOS RF front-end, highlighting the suitability of silicon photonics for future high-speed wireless fronthaul systems.

Index Terms—Silicon photonics, Mach–Zehnder modulator, Radio-over-Fiber, slow-wave electrodes, CMOS integration, mm-wave systems

I. INTRODUCTION

The rapid growth of data-intensive wireless services has driven the need for high-performance fronthaul solutions capable of supporting large bandwidths and dense antenna deployments. Conventional digital RoF architectures require high-speed data converters and digital processing at remote antenna units, resulting in increased power consumption and system complexity. Analog RoF addresses these challenges by directly transmitting RF signals over optical fiber, thereby centralizing processing and reducing remote unit hardware requirements.

Silicon photonics provides a scalable and cost-effective platform for implementing high-speed optical modulators due to its compatibility with standard CMOS fabrication processes. Among various modulator architectures, the Mach–Zehnder modulator (MZM) remains a preferred choice for analog applications due to its linearity and broadband capability. However, achieving mm-wave bandwidths in silicon MZMs requires careful optimization of phase shifters and electrode structures.

II. MODULATOR DESIGN AND IMPLEMENTATION

The proposed MZM employs high-speed PN-depletion phase shifters that utilize the plasma dispersion effect to achieve fast modulation. Device-level optimization is performed using TCAD simulations to extract carrier concentration profiles as a function of reverse-bias voltage. Optical mode simulations are used to compute effective index changes and optical loss, with phase shift efficiency estimated using established plasma dispersion models. A three-level doping

profile is adopted to balance modulation efficiency and free-carrier absorption loss.

To support wideband operation, a traveling-wave electrode structure incorporating inductive slot loading is implemented. This slow-wave electrode design enables simultaneous RF–optical velocity matching and impedance matching to CMOS drivers, which are critical for maintaining modulation efficiency across the full electrode length. The fabricated modulator has a total phase shifter length of 1.5 mm and supports differential drive operation for improved linearity and noise immunity.

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

Optical characterization confirms proper Mach–Zehnder interference with a measured free spectral range of approximately 22–25 nm, indicating good symmetry between the interferometer arms. Electrical measurements demonstrate an electrical–electrical 3 dB bandwidth of up to 38.8 GHz, validating the effectiveness of the slow-wave electrode design and RF–optical velocity matching.

To evaluate system-level performance, the silicon photonic MZM is integrated with a 16–32 GHz RF front-end implemented in 22 nm CMOS FD-SOI technology. The RF chain includes a cascode low-noise amplifier and stacked-FET driver stages capable of delivering large differential voltage swings to the modulator. Broadband interstage matching is achieved using magnetically coupled resonator techniques. The complete RoF receiver front-end demonstrates wideband operation with a peak input third-order intercept point (IIP3) of 1 dBm and a potential spurious-free dynamic range (SFDR) of 105 dB·Hz^{2/3}.

IV. CONCLUSION

This work demonstrates a 32 GHz differential silicon photonic Mach–Zehnder modulator optimized for analog Radio-over-Fiber applications. By combining PN-depletion phase shifters with slow-wave traveling-wave electrodes, wideband and CMOS-compatible operation is achieved. Integration with a high-frequency CMOS RF front-end validates the feasibility of silicon photonic–electronic co-design for next-generation wireless fronthaul systems.